APPENDIX 3 - LINKS TO 'MASTER' CONDITIONS AND OUTCOMES TABLES

Figure 1 is a snapshot of a 'master' table containing all conditions, challenges addressed, outcomes, and associated cases. It is the centerpiece of the qualitative content analysis stage.

[This is the link to the full 'master' table]

Case Name	Description	Condition Type	Condition Name	Challenge Addressed	Outcome Group	Specific Outcome Produced	Organization Type	Economic Sector	Location	Geographic Scope	Maturity
CoopCycle	Operates as federation of logistics cooperatives with board elected by cooperative members accountable to general assembly	Governance	Worker Ownership Structures	Gig economy platforms extracting value from workers	Community Value	Worker-owned alternative to exploitative platform labor models	Digital Cooperative	Federation of logistics messengers / cooperatives	France	Regional (Europe); Global North	Est. 2016; mature
CoopCycle	Evolved toward board of directors for daily operations while remaining accountable to general assembly, recognizing need for new structures when 'too big to be fully democratic'	Governance	Scale and Democratic Participation Navigation	Tension between operational efficiency and maintaining democratic participation as organization scales	Democratic Practices	Federated structure balancing efficiency with participatory governance	Digital Cooperative	Federation of logistics messengers / cooperatives	France	Regional (Europe); Global North	Est. 2016; mature
CoopCycle	Grew from 6 cooperatives in France to over 60 internationally while maintaining federated structure respecting each cooperative's autonomy	Governance	Federated Scaling Strategy	Tension between federation growth and maintaining individual cooperative autonomy	Impact	Transnational growth with locally-responsive governance	Digital Cooperative	Federation of logistics messengers / cooperatives	France	Regional (Europe); Global North	Est. 2016; mature
DAIR (Distributed AI Research Institute)	Operates as 'decentralized and resistant to hierarchical control' ensuring researchers retain autonomy and decision-making is distributed rather than concentrated in a single leadership body	Governance	Decentralized Decision-Making	Hierarchical control and concentrated decision-making in AI research	Democratic Practices	Researcher autonomy with distributed decision-making structure	Data Sovereignty and Digital Rights	Community centered AI research institute	USA	Global (distributed and embedded in communities)	Est. 2021; emerging
Detroit Community Technology Project (CTC)	Operates as 'a collective of local organizations' emphasizing 'collaborative and participatory governance' where 'the community itself is driving decisions'	Governance	Decentralized Decision-Making	Centralized top-down decision-making in technology projects	Democratic Practices	Community-driven collaborative decision-making mechanisms	Data Sovereignty and Digital Rights	Advocacy group for digital rights	USA	Built locally (in Detroit, Tennessee, LA, and NY) but now with a national span; principles the organization is built on and supports are aligned	Est. 2011; reasonably mature
Enspiral	Functions as open cooperative with collaborative governance, participatory funding, and self-managed teams	Governance	Decentralized Decision-Making	Hierarchical organizational control limiting participation	Democratic Practices	Self-managed collaborative structure distributing authority	Participatory Governance	Collaboration platform for mutual aid	New Zealand	Global	Est. 2010; semi-mature
First Nations Information Governance Centre (FNIGC)	Operates primarily within Canada while actively participating in global Indigenous data sovereignty networks through GIDA	Governance	Multi-Network Presence and Translocal Engagement	Tension between national focus and participation in global Indigenous sovereignty networks	Impact	Multi-scalar governance maintaining local control while contributing to transnational	Data Sovereignty and Digital Rights	Advocacy network - indigenous data rights	Canada	National, but participate in global networks; Global South	Est. 2010, but sovereignty work began in 1996; mature
Groupmuse	Converted to cooperative ownership ensuring revenues benefit community members rather than external investors	Governance	Community Ownership for Wealth Retention	Platform capitalism extracting value from cultural producers	Community Value	Cooperative structure retaining cultural value within community	Digital Cooperative	Community music events platform	USA	USA but slowly expanding - music is transnational; Global North	Est. 2013, coop conversion in 2020; reasonably mature
Guifi.net	Decentralized governance where local groups develop expansions independently while adhering to 'open, free, and neutral' principles through Foundation offering legal support	Governance	Federated Governance Models	Centralized telecommunications control limiting community autonomy	Balance	Federated structure combining local autonomy with shared principles	Peer Production	Commmunity wireless internet commons/internet infrastructure	Spain	National; Global North	Est. 2004; mature
Loomio	Operates with no traditional managerial hierarchy, relying on rotating coordination roles and using own deliberation tools for collective decision-making	Governance	Decentralized Decision-Making	Hierarchical organizational structures limiting democratic participation	Democratic Practices	Non-hierarchical cooperative governance with distributed coordination	Digital Cooperative	Software / Technology services	New Zealand	Global	Established 2011; mature, global technology services firm
Matrix	Creates decentralized, federated network where each homeserver is local in sense it can be privately run while remaining connected to global network	Governance	Federated Governance Models	Centralized communication platforms controlling user data	Balance	Federated architecture combining local sovereignty with global interoperability	Peer Production	Internet communications protocol	United Kingdom	Global	Est. 2014; mature
MiData	Enables formation of regional or national cooperatives sharing infrastructure while maintaining local governance	Governance	Community Ownership for Wealth Retention	Commercial platforms monetizing personal health data	Community Value	Cooperative data governance retaining benefits within member communities	Digital Cooperative	Data cooperative for health data	Switzerland	National; but permits global projects	Est. 2015; semi mature
MiData	Regional and national health data cooperatives sharing technical infrastructure while maintaining local governance autonomy, enabling global research collaboration under democratic data governance	Governance	Federated Democratic Governance	Tension between local data sovereignty and global research coordination needs	Balance	Interconnected cooperatives enabling cross-border research while preserving community	Digital Cooperative	Data cooperative for health data	Switzerland	National; but permits global projects	Est. 2015; semi mature

Figure 1 Representative sample of Master Table

Figure 2 is a snapshot of a matrix that plots Outcome Categories vs Condition Categories. The values in each of the cells represents the number of cases (from the 43 under study) that each intersection of outcome category and condition category have in common.

[This is the link to the quantitative matrix]

	TP1: Democratic Data Authority	TP2: Open Development and Transparency	TP3: Multi-Scale Architecture Design	TP4: Community-Centered Design Processes	TP5: Culturally-Grounded Development	TP6: Commons Resource Stewardship	RS1: Community Capacity Development	RS2: Access and Skills Democratization	RS3: Economic Value Redistribution
DP1: Decentralized Organizational Governance	0	2	1	2	0	0	2	1	3
DP2: Community Data Sovereignty and Control	6	2	1	0	0	0	1	0	0
DP3: Community-Controlled Infrastructure	0	0	2	0	0	0	0	2	1
DP4: Inclusive Participation and Representation	0	0	0	0	1	1	3	2	0
DP5: Open Access and Transparent Systems	0	4	2	1	0	1	1	0	1
DP6: Indigenous and Cultural Governance Integration	1	0	0	0	0	0	1	0	0
DP7: Legal and Institutional Protection	1	0	1	0	0	0	1	0	3
DP8: Resistance and Accountability Mechanisms	1	0	0	1	0	0	2	1	0
DP9: Local Capacity and Self-Governance	0	0	0	0	0	0	1	2	0
CV1: Worker and Member Economic Benefits	1	0	0	1	0	0	0	1	4

Figure 2 Representative sample of Outcomes x Conditions quantitative matrix

Figure 3 is a snapshot of a matrix that plots Outcome Categories vs Condition Categories. The values in each of the cells represents a qualitative analysis of the nature of the relationship between outcome category and condition category, as represented by the cases they have in common.

[This is the link to the qualitative matrix]

Outcome / Condition	TP1: Democratic Data Authority	TP2: Open Development and Transparency	TP3: Multi-Scale Architecture Design	TP4: Community-Centered Design Processes	TP5: Culturally-Grounded Development	TP6: Commons Resource Stewardship	RS1: Community Capacity Development	RS2: Access and Skills Democratization	RS3: Economic Value Redistribution
DP1: Decentralized Organizational Governance	—	enable participation	scale governance	facilitate deliberation	—	—	build agency	support practice	incentivize participation
DP2: Community Data Sovereignty and Control	establish control	enable scrutiny	enable practice	—	—	—	support practice	—	—
DP3: Community-Controlled Infrastructure	—	—	design scale	—	—	—	—	skill operation	retain value
DP4: Inclusive Participation and Representation	—	—	—	—	embed culture	enable practice	capacitate participation	democratize skills	—
DP5: Open Access and Transparent Systems	—	create transparency	scale access	operationalize participation	—	enable practice	build capacity	—	share value
DP6: Indigenous and Cultural Governance Integrat	ground data	—	—	—	—	—	support practice	—	—
DP7: Legal and Institutional Protection	protect data	—	enable practice	—	—	—	build defense	—	stake protection
DP8: Resistance and Accountability Mechanisms	enable practice	—	—	facilitate accountability	—	—	enable resistance	skill contestation	—
DP9: Local Capacity and Self-Governance	—	—	—	—	—	—	build local capacity	enable self-governance	—

Figure 3 Representative sample of Outcomes x Conditions qualitative matrix

Figure 4 and **Figure 5** are snapshots of the case positioning matrices (qualitative and quantitative), based on analysis of condition categories. In the figure below, the values in the cells are condition categories pertaining to the cases (rows) and condition types (columns). They are derived from the case equations (Appendix 4) and indicate what condition categories across the range of each condition type was present.

Case	Organization Type	Technology Practices	Resources	Governance	Technology Tools	Partners	Purpose	Organizational Information
FNIGC	Data Sovereignty and Digital Rights	TP1	RS1	GV4	TT3	—	PU1	—
GIDA	Data Sovereignty and Digital Rights	—	—	—	TT3	PT2	PU1	—
Masakhane	Data Sovereignty and Digital Rights	TP5	RS1, RS6	—	—	PT1, PT4	PU2	—
Te Mana Raraunga	Data Sovereignty and Digital Rights	—	—	—	—	PT2	PU1	—
DEF	Data Sovereignty and Digital Rights	TP6	RS2	—	TT2	PT1	—	OI5
Te Hiku Media	Data Sovereignty and Digital Rights	—	—	—	TT2	PT3	PU1, PU2	—
Detroit CTC	Data Sovereignty and Digital Rights	—	RS2	GV1	TT1	—	—	—
DAIR	Data Sovereignty and Digital Rights	TP4	RS1	GV1	—	—	—	—
Monlam AI	Data Sovereignty and Digital Rights	—	—	—	TT2	PT3	PU2	—
TIC A.C.	Data Sovereignty and Digital Rights	—	—	—	TT1	—	PU4, PU6	—
WAO	Data Sovereignty and Digital Rights	TP1	—	GV5	TT6	—	PU4	—
MAIAM NAYRI WINGARA	Data Sovereignty and Digital Rights	TP1	—	—	—	—	PU3	—
Guifi.net	Peer Production	TP3	RS3	GV3	TT1	—	—	OI2, OI3

Figure 4 Qualitative case positioning matrix

In the figure below, the scoring methodology described in Chapter 7 is applied to the qualitative case positioning matrix (above).

Case	Organization Type	Technology Practices	Resources	Governance	Technology Tools	Partners	Purpose	Organizational Information
FNIGC	Data Sovereignty and Digital Rights	0.75	1	0.25	0.5	N/A	1	N/A
GIDA	Data Sovereignty and Digital Rights	N/A	N/A	N/A	0.5	1	1	N/A
Masakhane	Data Sovereignty and Digital Rights	0.25	1	N/A	N/A	1	0.75	N/A
Te Mana Raraunga	Data Sovereignty and Digital Rights	N/A	N/A	N/A	N/A	1	1	N/A
DEF	Data Sovereignty and Digital Rights	0.25	0.5	N/A	0.5	0.75	N/A	0.25
Te Hiku Media	Data Sovereignty and Digital Rights	N/A	N/A	N/A	0.5	0.25	1	N/A
Detroit CTC	Data Sovereignty and Digital Rights	N/A	0.5	0.75	1	N/A	N/A	N/A
DAIR	Data Sovereignty and Digital Rights	0.25	1	0.75	N/A	N/A	N/A	N/A
Monlam AI	Data Sovereignty and Digital Rights	N/A	N/A	N/A	0.5	0.25	0.75	N/A
TIC A.C.	Data Sovereignty and Digital Rights	N/A	N/A	N/A	1	N/A	0.25	N/A
WAO	Data Sovereignty and Digital Rights	0.75	N/A	0	0	N/A	0.25	N/A
MAIAM NAYRI WINGARA	Data Sovereignty and Digital Rights	0.75	N/A	N/A	N/A	N/A	0.25	N/A
Guifi.net	Peer Production	1	1	0.75	1	N/A	N/A	0.75
Matrix	Peer Production	1	N/A	0.75	N/A	N/A	N/A	0.5

Figure 5 Quantitative case positioning matrix

[This is the link to the file with both case positioning matrices - conditions]

The table below is a condition category reference list, translating case positioning (from above) into implications for conditions.

Table 1 Condition category scores reference

Category	Score	Resonance	Cases	Key Necessary Conditions
TP1: Democratic Data Authority	+0.75	21	6	100% coverage for DP2
TP2: Open Development	+0.75	22	7	100% coverage for CV2
TP3: Multi-Scale Architecture	+1.0	24	6	83% coverage for BL1
TP4: Community-Centered Design	+0.25	8	2	—
TP5: Culturally-Grounded Development	+0.25	5	1	—
TP6: Commons Stewardship	+0.25	9	3	—
RS1: Community Capacity Development	+1.0	29	6	100% coverage for IM2
RS2: Access/Skills Democratization	+0.5	16	4	—
RS3: Economic Value Redistribution	+1.0	21	6	—
RS4: Alternative Economic Models	+0.5	10	3	—
RS5: Sustainable Resourcing	-0.25	6	2	—
RS6: Values-Aligned Allocation	0	9	2	—
GV1: Decentralized Authority	+0.75	21	6	100% coverage for DP1
GV2: Ownership-Based Control	+1.0	16	5	100% coverage for CV1
GV3: Federated Coordination	+0.75	14	3	100% coverage for BL1
GV4: Democratic Scaling	+0.25	10	2	—
GV5: Inclusive Expertise	0	7	2	—
GV6: External Democratization	+0.25	11	2	—
TT1: Physical Infrastructure	+1.0	21	5	100% coverage for DP3
TT2: Culturally-Grounded Artifacts	+0.5	14	4	—
TT3: Indigenous Data Infrastructure	+0.5	10	2	100% coverage for DP2, DP6, IM1
TT4: Participatory Platforms	+0.5	14	3	100% coverage for DP5
TT5: Digital Commons	+0.5	8	3	100% coverage for CV2
TT6: Algorithmic Transparency	0	7	2	—
TT7: Decentralized Architecture	+0.25	4	1	—
PT1: Reciprocal Knowledge Exchange	+0.75	21	6	100% coverage for BL3
PT2: Multi-Network Positioning	+1.0	24	6	100% coverage for BL2

PT3: Sovereignty Protection	+0.25	10	3	100% coverage for BL4
PT4: Expertise Integration	+0.5	17	4	—
PT5: Cross-Sector Collaboration	0	4	1	—
PU1: Indigenous Epistemology	+1.0	18	4	100% coverage for DP6
PU2: Cultural Preservation	+0.75	13	3	100% coverage for CV4, IM3
PU3: Value Reconceptualization	+0.25	6	3	—
PU4: Resistance-Construction	+0.25	7	2	—
PU5: Ecological/Social Transformation	+0.25	5	2	—
PU6: Democratic Citizenship	0	10	2	—
OI1: Worker/Community Ownership	+0.5	8	3	—
OI2: Decentralized Governance	+0.25	12	2	—
OI3: Anti-Extractive Circulation	+0.5	11	3	—
OI4: Commons-Based Production	+0.5	10	3	100% coverage for CV2
OI5: Local-Global Positioning	+0.25	12	3	—
OI6: Municipal/Public Governance	0	5	1	—
PO1: Policy Advocacy	+1.0	41	9	100% coverage for IM1
PO2: Resistance-Opposition	+0.25	9	2	100% coverage for DP4, DP8, IM1
PO3: Strategic Navigation	-0.25	8	2	—
RG1: Legal Innovation	+0.75	17	4	100% coverage for DP7
RG2: Ethical Frameworks	0	7	2	—
RG3: Strategic Regulatory Navigation	0	15	3	—
RG4: Self-Determined Sovereignty	+0.5	5	2	—
SC1: Inclusive Participation	+0.5	19	4	100% coverage for DP4
SC2: Movement Embeddedness	+0.25	7	2	—
SC3: Climate Response	0	9	2	—
PR1: Infrastructure Deficits	0	9	2	—
PR2: Colonial Knowledge Domination	+0.25	8	2	—
PR3: Cultural Erasure	+0.25	8	2	—
PR4: Algorithmic Opacity	+0.25	8	2	—

Figure 6 and **Figure 7** are snapshots of the case positioning matrices (qualitative and quantitative), based on analysis of outcome categories. In the figure below, the values in the cells are outcome categories pertaining to the cases (rows) and condition types (columns). They are derived from the case equations (Appendix 4) and indicate what condition categories across the range of each condition type was present.

Case	Organization Type	Democratic Practices	Community Value	Balance	Impact
DAIR	Data Sovereignty and Digital Rights	DP1, DP8	—	—	IM2
DEF	Data Sovereignty and Digital Rights	DP4	CV3, CV4	BL3	—
Detroit CTC	Data Sovereignty and Digital Rights	DP1, DP3, DP9	CV3	—	—
FNIGC	Data Sovereignty and Digital Rights	DP2, DP6, DP7	—	—	IM1, IM2, IM5
GIDA	Data Sovereignty and Digital Rights	DP2, DP6	—	BL2	IM1
MAIAM NAYRI WINGARA	Data Sovereignty and Digital Rights	DP2	CV4	—	—
Masakhane	Data Sovereignty and Digital Rights	DP4	CV4	BL3	IM2, IM3
Monlam AI	Data Sovereignty and Digital Rights	—	CV4	BL4	IM3
TIC A.C.	Data Sovereignty and Digital Rights	DP3, DP7	—	BL4	IM1
Te Hiku Media	Data Sovereignty and Digital Rights	DP6, DP7	CV4	BL4	IM3
Te Mana Raraunga	Data Sovereignty and Digital Rights	DP6	—	BL2	IM1
WAO	Data Sovereignty and Digital Rights	DP2, DP8	CV5	—	—
ARDC	Peer Production	—	CV3	BL3, BL5	—
Guifi.net	Peer Production	DP1, DP3, DP7	CV3	BL1	IM4

Figure 6 Qualitative case positioning matrix - outcomes

In the figure below, the scoring methodology described in Chapter 7 is applied to the qualitative case positioning matrix (above).

Case	Organization Type	Democratic Practices	Community Value	Balance	Impact	Total	Groups
Te Hiku Media	Data Sovereignty and Digital Rights	0.6	0.33	0.22	0.48	1.63	4
Masakhane	Data Sovereignty and Digital Rights	0.24	0.33	0.29	0.71	1.57	4
FNIGC	Data Sovereignty and Digital Rights	0.77	N/A	N/A	0.62	1.39	2
Detroit CTC	Data Sovereignty and Digital Rights	0.9	0.33	N/A	N/A	1.24	2
DEF	Data Sovereignty and Digital Rights	0.24	0.67	0.29	N/A	1.19	3
WAO	Data Sovereignty and Digital Rights	0.54	0.5	N/A	N/A	1.04	2
Monlam AI	Data Sovereignty and Digital Rights	N/A	0.33	0.22	0.48	1.03	3
GIDA	Data Sovereignty and Digital Rights	0.54	N/A	0.33	0.14	1.01	3
TIC A.C.	Data Sovereignty and Digital Rights	0.52	N/A	0.22	0.14	0.89	3
Te Mana Raraunga	Data Sovereignty and Digital Rights	0.36	N/A	0.33	0.14	0.83	3
DAIR	Data Sovereignty and Digital Rights	0.5	N/A	N/A	0.24	0.74	2
MAIAM NAYRI WINGARA	Data Sovereignty and Digital Rights	0.18	0.33	N/A	N/A	0.51	2
Guifi.net	Peer Production	0.67	0.33	0.33	0.48	1.81	4
Sensorica	Peer Production	0.14	0.72	0.5	N/A	1.37	3

Figure 7 Quantitative case positioning matrix - outcomes

[This is the link to the file with both case positioning matrices - outcomes]